Final Project Presentation: Guidelines

Course 22AIE437: Medical Image Processing

B.Tech CSE (AI) | Amrita School of Artificial Intelligence

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1. Format and Deliverables

This is a **group project**. Each group makes one presentation, one code submission and one report. In every place below, "you" means the group.

The presentation must be prepared in **LaTeX (Beamer)**. PowerPoint, Google Slides and PDFs exported from them are not accepted.

Submit the following:

- The **.tex** source file (with BibTeX file, if used)
- The compiled **.pdf** of the slides
- **The implementation code** (GitHub link or zipped folder) that reproduces the reported results
- **The project report** in Elsevier journal format (see Section 2A), submitted before the end semester examination
- **Group details:** group name or number, and the names and roll numbers of all members, on the title slide and on the report title page

Suggested length: 12 to 15 slides (about 10 minutes of presentation plus 5 minutes of questions).

2. Slide Structure

Section 1: Introduction (2 slides)

- **Slide 1: Problem and motivation.** State the medical problem, why it matters clinically, and why AI or image processing is needed.
- **Slide 2: Dataset and objectives.** Give the dataset name, number of images, classes or labels, and the train/validation/test split. State the task type (segmentation, classification or regression), your objectives, and the evaluation metrics.

Section 2: Base Paper and Replicated Results

- **Base paper details:** title, authors, journal or conference, and year.
- **Architecture:** include the architecture diagram of the base paper with a short explanation of its key blocks.
- **Results table:** show the results reported in the base paper and the results you obtained by replicating it, side by side.

Include the following according to your task type:

Task type	Required figures	Required table metrics (paper vs. replicated)
Segmentation	3 sample images, each shown as Input, Ground Truth and Your Predicted Mask	Dice and Jaccard (IoU)
Classification	Confusion matrix	Accuracy, Precision, Recall, F1-score
Regression	Scatter plot of predicted vs. actual values	RMSE and MAE

Section 3: Implementation Code with Results

- Bring your working code and run or demonstrate it. Show the code structure: data loading, model, training loop and evaluation.
- Include screenshots of training logs and curves (loss and metric) on the slides.
- Report the training settings: framework, optimizer, learning rate, batch size, number of epochs and hardware.
- Share the code as a GitHub link or a zipped folder. It must reproduce the reported results.

Section 4: Your Modification and Comparison

- **What you changed.** Describe the modification to the baseline architecture (for example a new block, attention module, loss function, activation function or augmentation) and show the modified architecture diagram.
- **Comparison table.** Compare the baseline (replicated) against your modified model using the same metrics as Section 2, on the same data split and settings.
- **Analysis.** Discuss the results in one of the two ways below.
  - **If results improved:** identify what contributed to the improvement. Support this with an ablation study (adding or removing one component at a time) and, where possible, qualitative examples.
  - **If results did not improve:** explain why. Possible reasons include small dataset size, overfitting, added complexity, a mismatch between the modification and the data, training instability or hyperparameter choices. State what you would try next.

An honest, well-reasoned analysis of a negative result earns full credit.

2A. Project Report (Elsevier Format)

In addition to the presentation, every group must submit a **project report in Elsevier journal format**. The report must be written in LaTeX using the Elsevier CAS single-column template:

Elsevier's CAS LaTeX Single Column Template (Overleaf)

- **Deadline:** the report must be submitted **before the end semester examination**.
- **Writing guidance:** the report must follow the guidelines given in the lecture presentation on how to write a good journal paper.

- Submit the compiled PDF together with the LaTeX source (.tex, .bib and figure files).
- The report should cover the same content as the presentation in full journal-article form: abstract, introduction, base paper and replication, methodology and modification, results and comparison, discussion, conclusion and references.
- Follow the template structure for title, authors, abstract, keywords, sections, figures, tables and the BibTeX reference style.
- List all group members as authors and include a short author contribution statement saying what each member did.

3. Suggested Slide Plan

Slides	Content	Section
1 to 2	Problem, motivation, dataset, objectives, metrics	Introduction
3 to 5	Base paper details, architecture diagram, key blocks	Base paper
6 to 7	Replicated vs. reported results table, task-specific figures	Base paper
8 to 9	Code structure, training settings, logs and curves	Implementation
10 to 11	Your modification and modified architecture diagram	Modification
12 to 14	Comparison table, ablation or failure analysis, qualitative results	Modification
15	Conclusion, future work, member contributions and references	Closing

4. General Rules

- **Figures:** use high-resolution images with readable labels, axes and legends. Every table must have a caption.
- **Fair comparison:** use identical data splits, random seeds and training settings for the baseline and the modified model.
- **References:** cite the base paper and any other sources. BibTeX is recommended.
- **Report:** the project report must follow the Elsevier CAS single-column format and be submitted before the end semester examination, following the lecture guidelines on writing a good journal paper.
- **Teamwork:** every member must present part of the talk and be ready to answer questions on any part of the project. Include a slide stating each member's contribution.
- **Originality:** the code and results must be your own.